From Distant Tales: Archaeology and Ethnohistory in the Highlands of Sumatra

Edited by

Dominik Bonatz, John Miksic, J. David Neidel,
Mai Lin Tjoa-Bonatz

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THE NEOLITHIC IN THE HIGHLANDS OF SUMATRA: PROBLEMS OF DEFINITION

DOMINIK BONATZ

Introduction

Archaeologists use the terms Paleolithic and Neolithic to refer to the earlier and later epochs of human prehistory, which can be distinguished by the types of artefacts people left behind, especially the types of stone tools they used. These periods correspond with two geological phases: the Pleistocene and the Holocene. The earliest part of the Holocene, which began about 11,500 years ago and which marks the official end of the last ice age, coincides closely with the first human experiments in the domestication of plants. The transition to agriculture marks a radical change in the development of human society and is therefore often labelled as the Neolithic Revolution (Manzanilla 1987, Price/Gebauer 1995). For V. Gordon Childe, who coined the term in the 1920s, the Neolithic Revolution was mainly an economic transformation which resulted in a fundamental shift in orientation towards stable food production strategies and their accompanying techniques of food storage, in contrast to earlier subsistence-based economies that relied wholly on foraging. Since then, archaeologists have used the term "Neolithic" in a broader sense to refer to periods of human civilization, in which farming was the main method of subsistence, sedentism was more widespread than seasonal mobility, and people relied on pottery vessels for the preparation and consumption of food, as well as on ground stone tools.

The phrase Neolithic Revolution is still widely used by archaeologists, even though most of them would nowadays argue that the development to agriculture and sedentary life should be seen as gradual evolutionary process rather than a sudden cataclysm. This process started independently in different parts of the world, and it entailed major changes in the structure

and organization of societies as well as a totally new relationship with the environment. Attempts have been made to identify zones of "pristine domestication" in the Near East, northern Africa, Mesoamerica, North America, and northern China (Minnis 1985; Smith 1989; Harlan 1995). However, the question of the spread of agriculture and animal husbandry beyond these zones is important as well. In addressing the issue, how and why the formation of Neolithic societies took place in so many places and under such different conditions, a variety of premises must be made. First, it is important to appreciate that the complex features associated with Neolithic societies such as farming, pottery production, and ground stone tools varied by region and developed at different rates. Second, changing environmental conditions and increasing social interactions do not inevitably change the economic orientation of human populations. It is therefore reasonable to question the teleological character of agriculture and sedentism which is often taken for granted. Third and finally, archaeological data are not necessarily indisputable to our interpretation of Neolithic communities. They are just means to generate and to test our set of hypothesis and reconcile often contradictory evidence.

The process by which specific Neolithic communities could have emerged in island Southeast Asia will now be modelled in the case of Sumatra. With special focus on the highlands, the available archaeological data will be examined in order to define the distinct nature of a Neolithic period. However, Sumatra neither belonged to the zones of "pristine domestication" nor was it geographically directly connected to one of them. It is therefore questionable if the development of sedentary agricultural communities was an inevitable step in the prehistory of the island. Since alternative options to agriculture and sedentism may have existed for a long time up to the present, the possible causes of food production must be discussed on a theoretical level. For this purpose, different models which seek to explain the transition to agriculture are presented in the following section. At the end of this paper, the model which most closely fits the archaeological data is taken to formulate a hypothesis on the spread of Neolithic communities in the highlands of Sumatra.

Theoretical Approaches on the Origin of Agriculture

In her article on the "Origins of Food Production in the New World" Barbara Stark (1986) describes three main types of models which are used by archaeologists to explain the transition to food production: "push" models, "pull" models, and "social" models (see also Bogucki 1999: 847-848, fig. 21.4). "Push" models were proposed as early as the late 1960s

(e.g. Binford 1968; Flannery 1969). They claim that stress situations caused by exceeding resources due to population growth or climatic changes have pushed people to adopt agriculture. Such models have been common to explain the origins of agriculture in the Near East. For K. V. Flannery (1969), for example, population pressure was the main cause for people to move from the optimal zones of the Fertile Crescent to adjacent marginal zones where they started to cultivate cereals for the first time, that is about 10,000 years ago. With new evidence at hand, such as pollen diagrams which clearly showed that the earliest cultivating communities were situated within the natural habitat of wild cereals (van Zeist/Bottema 1991), later developed "push" models tended to prefer multiple factors for the transition to food production including changing environments, alternating subsistence strategies, and social organization (e.g. Moore 1982, 1989). Most recent finds in the Upper Euphrates region of modern Turkey prove the outstanding complexity and economic success of early Neolithic hunter-gatherer societies which even managed the construction of monumental cultic building complexes (Schmidt 2006). This evidence once again raises the question why hunter-gatherers found that the advantages of an economic system involving food production outweighed the options available to them as foragers. Some archaeologist consequently started to argue that cultural and symbolic processes are not secondary or tertiary in relation to the economic and social spheres. In fact these processes are considered as one of the main reasons to push people into agriculture (e.g. Hodder 1987; Cauvin 1994).

In contrast to "push" models "pull" models do not concentrate on single stress factors but emphasize the role of increasing reliance on specific resources. The continuous exploitation of specific plants and animals led to the dependence on them. Modifications of plant-human-animal relationships therefore pulled people into agriculture. Such models were first proposed for the Mesoamerican highlands (Flannery 1968). Later they were also used to explain the origins of agriculture in the Near East, especially the Levant (Henry 1989; Bar-Yosef 1995: 65-72). Here the emergence of farming communities is seen as a socio-economic response to the forcing effects of climatic changes. The first change occurred around 12,500 years ago, in which an increase in temperature promoted a complex foraging system which heavily relied on wild cereals. This system collapsed after a second climatic change about 2,000 years later. As an option to balance the lack of wild plant resources people began to cultivate cereals in vegetationally rich areas with high water tables such as the Jordan valley, the lakeshores or riverbanks of the so-called Levantine Corridor (Bar-Yosef 1995: 70, fig. 3.4).

Finally, "social" models stress the importance of collective food consumption for the development of food production. As ethnographic examples from foraging and cultivating societies show, many aspects of status depend on the provision of food for feasting. Therefore, the demand for food to meet social needs may have strengthened efforts to maximize food supply. Approaches to identify the social power of feasting as one cause for food production are, however, rare since the archaeological evidence is often elusive. B. Hayden (1992) has proposed such a model which he terms the "competitive feasting" model. He argues that the need to generate large amounts of desirable foods in order to stage competitive feasts would have stimulated foragers to experiment on cultivation. This hypothesis has no specific geographic reference but it is in line, for example, with observation from rice cultivating societies of East and Southeast Asia where surplus production apparently led to status and wealth (Higham 1995: 147-148). A similar approach relates surplus production to trade. C. Runnel and Tj. van Andel (1988) propose that agriculture developed in the eastern Mediterranean because a surplus of food was needed for trade or to support craftsmen who made goods for trade.

The link between agriculture and sedentism is implicit to most models which aim to explain the transition to food production. This is especially true for theories on the spread of Austronesian-speaking groups into the Indonesian archipelago and Oceania. These groups are believed to have been farmers who cultivated rice and other crops, and who brought with them material innovations such as pottery and new lithic industries. These early farming communities would then meet the characteristics of a period which is called the Neolithic. The conception of the term sedentism, however, needs special reflection, and the material objects which could trace Neolithic communities still need to be identified. Before summing up the archaeological data which relate to these critical points of interpretation, a short review on the research history will be given.

The Neolithic in the Southeast Asian Archipelago: The Question of Early Farming Communities

In his pioneering work "The Stone Age of Indonesia" (1972), Hendrik Robert van Heekeren puts the beginning of Neolithic period after the Mesolithic or Sub-Neolithic stage some 10,000 years or more ago. He concluded that for most parts of Indonesia archaeological evidence for the Neolithic was insufficient (van Heekeren 1972: 203). Only two artefact groups, rectangular polished adzes and paddle-and-anvil pottery, were

considered by him as indicative of the Neolithic. However, only a few sites in the northern and eastern Indonesian archipelago had yielded pottery within a distinct Neolithic context, and most of the stone adzes (fig. 3-5) were recorded without any archaeological context at all.[1]

When new archaeological material became available in the 1970s and 1980s, the efforts to define a Neolithic period shifted to the question of the origins and dispersal of farming communities, and the archaeological data to which agricultural activities could be attributed. The archaeological assemblages which Peter Bellwood in his research on the expansion of Austronesian-speaking peoples associates with agriculture are pottery, polished adzes, ornaments of stone and shell, spindle whorls, and village-sized settlements (Bellwood 2004: 21, fig. 7.12; 2006: 106). The documentation of at least some of these assemblages at archaeological sites in the Philippines, Malaysian and Indonesian Borneo, Sulawesi, eastern Java, and the northern Moluccas, is taken as proof of the Austronesian colonization, and with it the spread of farming communities, during the second and early first millennium BCE.

However, unequivocal proof of agricultural activities is rarely given. In one site, Gua Sireh cave in northwest Sarawak, rice phytoliths were recorded in a context of around 2000 BCE (Bellwood et al. 1992; Beavitt et al. 1996). In another, Kimanis Cave in East Kalimantan, some potsherds were found that seemed to contain rice impressions (Bellwood 2006: 111). A date of 2000 BCE has been proposed for rice grains from a Maros cave in South Sulawesi (Paz 2005). However, secure evidence for rice cultivation in South Sulawesi is not before approximately 500 AD when rice still was produced on a modest scale (Bulbeck/Caldwell 2008: 15). The absence of any fragments of processed or unprocessed cereal, such as *Orzya spp*. (rice), at sites where recent research, including archaeobiological studies, has taken place, however, is even more striking (Barker et al. 2002: 160-161).[2] Bellwood, nevertheless, believes that rice cultivation spread from the Philippines through Borneo into western Indonesia after 2,500 BCE connecting it with the dispersal of paddle impressed pottery style (Bellwood 2006: 111). He distinguishes this movement from that which carried red-slipped pottery into the eastern

[1] The illustrated examples on fig. 3-5 are surface finds found in the vicinity of the excavations at Muara Payang in Sumatra, see below.

[2] In other cases the evidence is just too scarce. At the west mouth of Niah Caves in Sarawak, for example, a single rice grain found in a potsherd associated with a burial yielded a date of c. 3,000 BCE (Brooks et al. 1977). Since this remained a unique find, the present excavators do not believe that the people were already cultivating rice (Doherty et al. 2000).

Indonesian archipelago and Oceania where the farmers apparently replaced rice cultivation by tubers and fruits.[3]

Bellwood's Austronesian dispersal hypothesis has attracted much attention and, with some modifications, seems to be accepted by many researchers (Bellwood/Renfrew 2002; Forestier 2003). It mainly works however for the northern sphere of the Southeast Asian archipelago, but does not provide us with a model for the spread of agricultural communities into the western part of the Indo-Malaysian archipelago (e.g. Forestier 2003: 32, fig. 3). Due to the lack of archaeological information, the latter migration remains largely theoretical. The date for the assumed arrival of Austronesian-speaking populations into Sumatra is for example considered to be quite recent at around 500 BCE (Miksic 2004: 191), and it cannot therefore be reliably stated that the beginnings of food production in this area are related to the arrival of external language groups.

The challenge for archaeologists, then, is to understand why and when the transition from foraging to farming, i.e., the process of Neolithisation, took place in Sumatra. Putting this question into a framework of current archaeological research, I will now review the evidence so far at hand, starting with Sumatra in general and then focussing specially on the highland's region.[4]

The Neolithic in the Prehistory of Sumatra: Current Stage of Research

Fundamental to the definition of a Neolithic period in the prehistory of Sumatra is recognition of the preceding period. So far, the only recognized post-Pleistocene, pre-Neolithic tradition in Southeast Asia is the Hoabinhian (Glover 1977). The term Hoabinhian, borrowed from excavations in the

[3] Note, however, that red-slipped pottery has recently been found at Pondok Silabe in Sumatra (see below).

[4] This chapter draws upon the results of several archaeological investigations carried out in Sumatra during the last 20 or so years including our own archaeological project in the highlands of Jambi. This project was initiated in 2003 as a joint research project between the National Research and Development Centre of Archaeology (Puslitbang Arkenas) in Jakarta and the Institute of Ancient Near Eastern Archaeology at the Free University Berlin and was sponsored by the Swiss-Liechtenstein Foundation for Archaeological Research Abroad in Zurich. Special credit is given to those team members who have continuously contributed to the progress of the project, especially J. David Neidel (National University of Singapore), Bagyo Prasetyo (Puslitbang Arkenas, Jakarta), Mai Lin Tjoa-Bonatz (Berlin), Tri Mahaini S. Budisantosa (Balai Arkeologi Palembang), and Agus Widiatmoko (Archaeological Preservation Office Suaka, Jambi).

North Vietnamese province of Hoa Binh, refers to a "technocomplex", rather than a culture, characterized by the ample use of pebble artefacts, especially simple pebble tools, unifacially retouched pebble tools ("Sumatraliths"), edge ground tools and short axes (Moser 2001: 30-33). Hoabinhian tools start to appear c. 14,000 BCE in Vietnam and thereafter spread over mainland Southeast Asia to the Philippines, Borneo, and Sumatra. The end of the Hoabinhian period depends much on local developments and can therefore only be dated with much variation to between the 9th and 4th millennium BCE or even much later.

Some scholars have compared the Hoabinhian with the idea of the Mesolithic in Europe (Narr 1966; Gorman 1969; van Heekeren 1972). This comparison, however, is inappropriate since neither the lithic tool kits nor the ecological-economic circumstances of the Hoabinhian can be paralleled with the Mesolithic (Moser 2001: 30). Others, therefore, have reasonably questioned the applicability of chronological phases such as Mesolithic and Neolithic for Southeast Asia (Callenfells 1936; Hutterer 1985). Efforts to redefine the archaeological chronology of Southeast Asia are nonetheless limited to the point that a common pattern for periodisation can not be found within an area of such geographic and biodynamic diversity. A good example for this dilemma is the relation between the Hoabinhian and the Neolithic in Sumatra.

In Sumatra, Hoabinhian sites usually refer to large shell middens (or kitchen middens) known from coastal areas in the north between Aceh and Medan (Brandt 1976; Edwards McKinnon 1990; Moser 2001: 130-131). One of the rare excavations at such an open air Hoabinhian site in Sukajadi near Medan has yielded a non-calibrated radiocarbon date of 7,340 ± 360 BP (Bronson/Glover 1984). Only recently, two Hoabinhian cave occupations have been excavated by a French-Indonesian team at the site of Tögi Ndrawa on Nias Island (Forestier et al. 2005) and at Gua Pandang near Padang Bindu in South Sumatra (Forestier et al. 2006: 183-185). In addition to the typical pebble artefacts, 11 radiocarbon dates from the shell accumulation inside the cave of Tögi Ndrawa provide us with one of the longest stratigraphic sequences in an Indonesian cave site dating from about 12,000 until 1,300 BP, and thus prove the persistence of the Hoabinhian tradition in such remote places as the island of Nias. Dates from Gua Pandang range between 9,270 until 6,590 BP. Findings from these site are especially important since they prove the expansion of the Hoabinhian technocomplex through Sumatra.

Two single finds from our own survey in the highlands of Jambi can now be added to this information. One is a Hoabinhian pebble tool found in the Air Hitam valley south of Lake Kerinci (Bonatz 2006: 317,

fig. 29.10), the other is a bifacially retouched tool, apparently an adze, collected from the surface at Bukit Arat in the Serampas valley (fig. 3-1 and see below). In comparison to similar tools, e.g. from the Niah Cave (Moser 2001: pl. liv), this tool can be considered to belong to the late or post-Hoabinhian technocomplex. Both finds, the pebble tool and the adze, raise the possibility that Hoabinhian technology also spread througout the highlands.

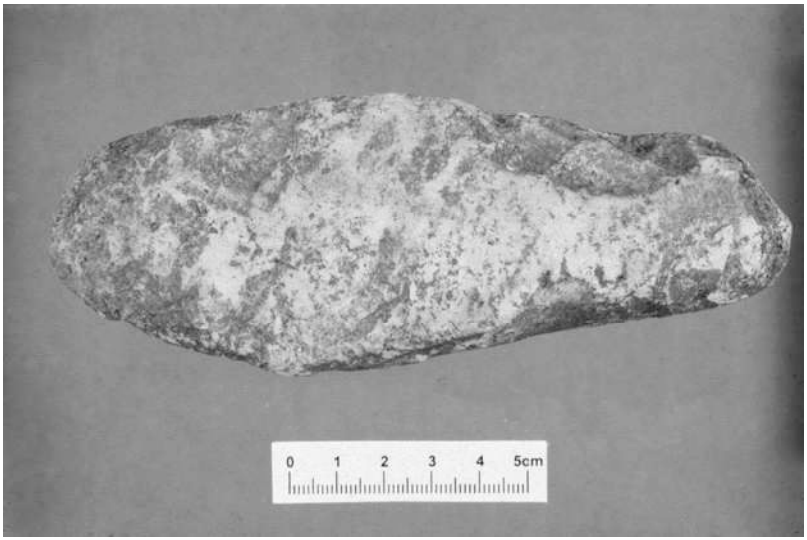

Fig. 3-1: Late or post-Hoabinhian adze from Bukit Arat, Serampas

The peoples using Hoabinhian tools are believed to have been hunter-gatherers since no Hoabinhian site has provided undisputed evidence of domesticated plant remains or bones of domesticated animals. The site of Tögi Ndrawa, for example, shows a spectrum of food typical for the Holocene local natural environment: wild-boar, deer, monkey, fish, seafood, and shell (*Thiaridae brotia, Veneridae venus*) (Forestier et al. 2005). Thus one could propose that the transition to food production marks the end of the Hoabinhian tradition, and with that the beginning of the Neolithic period. Archaeological research in other parts of Southeast Asia, however, demonstrates that foraging traditions must not have necessarily been replaced by farming. T. N. Headland and L. A. Reid (1989), for example, propose that in northern Luzon in the Philippines rice-farming populations and *Negrito* hunters were living in close vicinity

for at least the last 3000 years. The interaction with the Austronesian speaking farmers would have been so intensive that the *Negritos* adopted the language as their own (Headland/Reid 1989: 46). But they did not take over agriculture. One should therefore be cautious in putting periodical terms such as Hoabinhian, Mesolithic or Neolithic in a strict unilinear evolutionary and chronological relation. As archaeological terms they rather serve to set up criteria for the distinction of the variety of socio-economic orientations in human civilizations and the identification of their material correlates.

The main problem of inquiry, then, is how to define a Neolithic assemblage which seems to have replaced the Hoabinhian technocomplex if, on the one hand, no criteria for its classification exist, and, on the other hand, no additional information stems from archaeobiological or archaeozoological studies. To focus on this problem, two sites at the border of the Sumatran highland region will briefly be compared, before moving on to examine the highlands themselves.

The first example, Tianko Panjang Cave, is situated at the distal end of a limestone promontory that borders the valley of the Tianko River. Immediately south of Tianko is a wide break in the northeastern wall of the Barisan Mountains through which the Mesumai, Merangin, and Tembesi rivers drain the actively volcanic region around Kerinci. After a short test excavation by J. Zwierzycki in the 1920s, the site was extensively excavated by a joint American-Indonesian team under the direction of Bennet Bronson and Teguh Asmar in 1974 (Bronson/Teguh Asmar 1975).

The soil in a sheltered area of the cave was excavated to 2 m deep. Pottery was only found in the upper 40 cm, and thus appears only later in the cave's occupational period. The potsherds were found mixed with obsidian blades, and obsidian blades continued to be the major find group down to the bottom of the excavations. Among the 600 plus pieces collected, a very low percentage of intentionally shaped and retouched artefacts were recognized. As formal tool types such as shouldered scrapers, borers, gravers, and arrowheads did not occur, the excavators made the still valid conclusion that taxonomic procedures for lithic artefacts will not work in central Sumatra.

Unfortunately, no radiocarbon dates are given for the upper strata in Tianko Panjang Cave, and therefore do not cover those layers that contained pottery. The three samples taken are all from layers deeper than 90 cm. They provided dates between c. 9,210 and 10,250 BP (Bronson/ Teguh Asmar 1975: 136), and are thus comparable to other middle Hoabinhian sites in Sumatra and on the Southeast Asian mainland

(especially Thailand). The problem, however, is that during the apparently long period of occupation at Tianko Panjang, no change in the use and in the shape of lithic industries was observed, while the appearance of pottery seems to be abrupt and again without a secured date. No polished adzes or other lithic artefacts typical for a Neolithic site were found, and furthermore no faunal or plant remains are recorded which could lead us to surmise that the cave-dwellers had at some point experimented with food production. In contrast, they are believed to have been successful enough at exploiting wild forest produce to have resisted true agriculture until long after other peoples of the region had fully entered the Neolithic (Bronson/ Teguh Asmar 1975: 132).

In conclusion, Tianko Panjang can be characterised as a "negative" example of a site which could for a very long time be an attractive dwelling place for foraging people who from a certain moment on incorporated pottery in their lifestyle but who never experienced the traditional Neolithic achievements of agriculture and animal husbandry.

Following Tianko Panjang, the second example is Pondok Selabe I which is situated in the karst region of Batu Raja near Padang Bindu in South Sumatra. Recent excavations conducted by the IRD (Institut de Recherche pour le Développement) and the Puslitbang (National Research and Development Center of Archaeology) in Jakarta have unearthed a cave site in which three different phases of occupation were clearly documented and fortunately dated by means of radiocarbon analysis (Simanjuntak/Forestier 2004; Forestier et al. 2005; 2006: 185-187). The sequence starts with a "Metallic Phase" in the upper stratum of the cave around 1,800 BP, and ends with an "aceramic Phase" in the deepest stratum, dated between 3,000 and 4,500 BP. This earliest phase yielded no pottery but in common with Tiangko Panjang did contain lithic blade implements. Based on a single radiocarbon sample, the intermediate stratum is dated to c. 2,700 BP. This phase is labelled as Neolithic since it contained potsherds incorporating regional traits of Neolithic pottery that while not known in Sumatra are typical for Neolithic sites in the Philippines, northern Borneo, Sulawesi, the Moluccas, and East Java: black and red slipped pottery, a pot with a cord decoration, and potsherds with deep incisions.[5] All of the pottery is thin-walled and high fired. This pottery is really quite remarkable and raises the question of Neolithic farming communities coming from the north or the east coast, and bringing with them new technologies such as pottery making. If these

[5] Personal communication with Hubert Forestier and Bagyo Prasetyo; see also Truman Simanjuntak et al. (2005: 49) with illustrations of impressed pottery.

farmers are the same who brought Austronesian languages to Sumatra then a new and very important argument can be made to stress the theory of a western route of the Austronesian expansion. For the moment however all that is speculation and it would seem more appropriate to start with a discussion about the nature of the Pondok Selabe site.

Here the excavators have made a convincing suggestion (Simanjuntak et al. 2005: 49). The invention of pottery must not necessarily reflect a change in population, or a change in living circumstances of those who once settled within the cave. The absence of other typical Neolithic artefacts like the polished stone adze combined with the presence of bones from wild boar, deer, and monkeys, instead of domesticates again, like Tianko Panjang, speaks in favour of a group of hunter-gatherers who made seasonal use of the cave. The pottery then could have been an exchange good received from early farmers or horticulturalists who settled in the open terrain not far from the caves.

In general, the equation between pottery and agriculture, and thus with the Neolithic, must be questioned. In many cultural contexts it is well attested that both do not necessarily relate to each other (c.f. Rice 1999). It is, for example likely that pottery was invented in northern China prior to the beginning of plant cultivation (Higham 1995: 133; Guo/Li 2000). The complex long standing hunter-gatherer cultures of Jomon in Japan and Chulmun in Korea employed pottery for thousands of years before they started to cultivate plants (Underhill/Habu 2005: 134-141). Since they also have used polished stone axes and practiced sedentary life both have been equated with Neolithic cultures in Europe, China, and elsewhere (Underhill/Habu 2005: 135). In contrast, the early Neolithic communities of the Near East had experimented for about 3000 years with techniques of food production before they started to use pottery for storage and food consumption (e.g. Thissen 2007).

The problem, then, for archaeologists is how to tease out early farming groups in prehistoric environments such as Sumatra, and how to find their settlement places given that the evidence we have is often ambiguous. We see that in places where pottery has been found life and subsistence continued on Sumatra without drastic changes. Therefore, first the context and the combination of various artefacts have to be investigated before any conclusions can be made about the residential nature of a given site.

Keeping this difficult starting point in mind, I will now shift to the highlands and consider the three essential types of archaeological sites so far investigated there: open air sites, burial sites, and megalithic sites. After that a second important source for archaeological information will be considered: the natural environment.

Archaeological Sites in the Highlands of Sumatra

Open air sites

Much of prehistoric research depends upon the visibility of sites to archaeologists. In the landscapes of island Southeast Asia especially, open air sites are hard to identify because environmental conditions and perishable building materials such as wood and bamboo did not allow for the accumulation of cultural deposits, and the formation of mounds which could be explored by means of stratigraphic excavations. The typical situation, instead, is that which we found when excavating the site of Sungai Hangat in the highlands of Jambi in 2003 (Bonatz 2004: 118-123; 2006: 313-316; Bonatz et al. 2006: 495-496). During excavations many potsherds including Chinese porcelain from the Song Dynasty (1127-1279) until the Qing Dynasty (1644-1912) were collected, but since virtually none of the pieces was found in a stratified context, no chronological relationship could be established between the datable imports and the local earthenware or other finds like metals and stone implements. Most of the other sites we surveyed in the highlands of Jambi have the same problem: Different artefacts of different periods appear within the same context, thus the criteria to classify a site and its find assemblage as typical for a certain period are lacking.

An exception might be the sight of Bukit Arat which has been identified in 2006 in close vicinity to Renah Kemumu in Serampas.[6] It is a small mound situated at the boarder of the rice fields from where it overlooks a good part of the valley. A survey of the place provided surface finds of the late Hoabinhian tool, already mentioned here (fig. 3-1), and pieces of obsidian. Due to the importance of these finds large-scale excavations were carried out in 2008. The excavated surface of about 250 m^2 yielded a huge amount of potsherds and obsidians, some flints and a set of round grinding stones (fig. 3-2). The obsidians include totals of 1178 flakes, 220 blades, 18 roughly ovoid hammers for producing the flakes and blades, about 200 pieces of core preparation debris, and 1392 chips. Thus the site exhibits all characteristics of an obsidian workshop. But it provides also evidence for domestic activities. 1040 pieces of earthenware were collected among which 55 are rims. The fabric of the potsherds is almost exclusively coarse red with many thin-walled pieces. Some show traces of paddle-marked decorations. The rims mostly stem from cooking pots but also few bowls and medium-sized containers are attested.

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[6] For a map of the region see Neidel, infra: fig. 13-1.For a preliminary report on the excavations at Bukit Arat see Bonatz 2009.

Fig. 3-2: Find assemblage including potsherds, obsidian blades and hammers, and stone tools from test pit 2 in Bukit Arat, Serampas

The finds from Bukit Arat form an exceptionally homogeneous assemblage which, in theory, shows all the characteristics of a Neolithic assemblage. One might therefore be tempted to speculate about the nature of the site as one which marks the beginning of farming and sedentism in the Serampas valley. In order to achieve an idea of the age of the site, potsherds from different find contexts were analysed by means of thermoluminescence. They lead to surprisingly high dates between c. 1650-800 BCE.[7] This date is among the earliest ever since achieved for a Neolithic site in Sumatra. It can only be compared with another recently excavated open air site at Benua Keling Lama on the Pasemah plateau, which has a "Neolithic" level dating back to c. 1550 BCE (Simanjuntak et al. 2006; Guillaud et al., infra). This level contained the fragment of a

[7] The samples were analysed by CUDaM, Laboratorio di Termoluminescenza, Università degli Studi di Milano-Bicocca, using the standard *fine-grain* technique: BA 08-57-2: 1205 ± 270 BCE (Lab. Code D2242), BA 08-57-4: 1285 ± 265 BCE (Lab. Code D2241), BA 08-107-1: 910 ± 200 BCE (Lab. Code D2243), BA 08-123-4: 1340 ± 250 BCE (Lab. Code D2245), BA 08-157-2: 1390 ± 260 BCE (Lab. Code D2246), BA 08-157-8: 1040 ± 235 BCE (Lab. Code D2247).

polished stone tool. Considering the fact that the dates for the Neolithic in Pasemah and now also in Serampas are older than any date so far obtained from a site at the foothills or in the lowlands, one might be tempted to locate the beginnings of the process of Neolithisation in the highland regions (c.f. Guillaud et al., infra).

Burial sites

In other places of Southeast Asia burial sites are by far the most promising sites to identify Neolithic assemblages. One need only recall the cave burial sites of Gua Cha in the Malaysian Peninsular and the Niah caves in Sarawak, Borneo. The Gua Cha burials date between 2,500 and 1,000 BCE. Most of the human skeleton at that site were found in extended positions, and were buried with grave goods that included stone bracelets, quadrangular-sectioned adzes, and pottery of footed, round, and flat-based forms with a predominance of cord-marked decoration (Bellwood 1997: 260-265). In the Neolithic, between about 3,000 BCE and 800 CE, the Niah Cave complex was exclusively used as a cemetery. The broad sequence of burial types includes inhumations in extended and flexed position, cremations, and jar burials in lidded jars (Bellwood 1997: 238-241; Barker et al. 2002: 149, 159-160). Typical artefacts of the Neolithic assemblage at the Niah caves are polished quadrangular and trapezoidal stone adzes, paddle-decorated types of earthenware, and the double-spouted earthenware vessels, the so-called Niah Cave vessels.

For the highlands of Sumatra the main problem in finding burial places stems from unfavorable conditions for the preservation of human skeleton material. In many parts of the region the soils are acidic, leading to poor preservation of calcium, so that all bone material would quickly have been destroyed. There is, however, more than one way to identify a burial place. One distinct practice of burial, for example, is the deposition of the human corpse or cremation ash into a jar (or urn), which is quite a common practice in prehistoric burial sites of the Indo-Malayan archipelago. Among the burial types attested in the Niah Caves, for example, are also lidded burial jars. While the Niah Cave complex is a distinct Neolithic burial site, the bulk of the jar burial sites in Indonesia, are believed to coincide with the Metal period (Bellwood 1997: 306-307).

Fig. 3-3: Jar burials at Muara Payang, Lahat (printed with courtesy of Balai Arkeologi Palembang)

Fig. 3-4: Presumed jar burials at Renah Kemumu, Serampas

At the first glance this seems to be also true for jar burial sites which have more recently been excavated by Indonesian archaeologists in the mountainous region west of Lahat in South Sumatra. The burial grounds at Muara Payang (fig. 3-3; Kristantina Indriastuti 2002, 2003) and Muara Betung (Retno Purwanti 1997, 2002; Sri Mulyati et al. 1999) are marked by unworked stones (so called dolmens). They contain classical vertical mouth-to-mouth jar burials meaning that one pot has been used as a cover or lid for the other pot.[8] Human skeleton material inside the jars is very rare but in Muara Betung the teeth and fragments of a human skull have been found in one of the jars. These interments are, however, remarkable for the appearance of skeletons in extended position which were placed beside the jars. Thus, we are dealing either with different burial rites or one rite which includes a primary and a secondary burial. Unfortunately, no analysis of bones or soil samples has yet been made. Absolute dates for these sites are therefore unavailable. The ceramic assemblage which was found in the area of the burial grounds is also not very helpful for precise dating. It includes Chinese porcelain which ranges from the Song until Qing period, i.e., from the 11th century onwards.

Another jar burial complex was excavated during our 2005 field campaign in Renah Kemumu in Serampas (fig. 3-4; see also Bonatz et al. 2006: 500-502, fig. 10-11), which lies only a short distance from the Bukit Arat site. I have to admit that we cannot claim to be absolutely sure of our interpretation of Renah Kemumu as a burial ground, since no traces of human bones could be identified even with the help of chemical analysis. The jars, of which 35 have been mapped and four excavated, range from 0.60-1.20 m in diameter. They have no visible lids, thus the question how they were covered remains open. Among the body and rim sherds fallen into the jar there are also sherds of much smaller vessels which could be interpreted as funerary gifts. Conventional radiocarbon dating of a charcoal sample taken from the fill of one jar yielded a date of 1100 ± 120 BP.[9] In addition, optically-stimulated luminescence (OSL) measurement of one potsherd taken from the body of a jar dated this piece to 813 ± 128 CE, and a second piece from one of the smaller vessels to 1039 ± 124 CE.[10] Sufficient evidence exists, therefore, to conclude that the

[8] For similar jar burial sites in the coastal area see Padang Sepan north of Bengkulu (Kristantina Indriastuti 2002a, 2003a, 2004) and Lebakbandung near Jambi (Eddy Sunarto et al. 1996).

[9] Non-calibrated radiocarbon date, without Geolab-number Radiocarbon Dating Laboratory Bandung

[10] The analysis of the potsherd samples was conducted by C. Goedicke of the Rathgen Laboratory at the Staatliche Museen in Berlin, using the single grain OSL system.

burial ground at Renah Kemumu was used between 800 and 1,100 CE. This, however, is apparently not the date which is commonly associated with a Neolithic burial site. Instead it falls within the range of other jar burial sites in Indonesia and those from South Sumatra which certainly do not date back to before the first millennium CE and which are commonly associated with the Metal period.

One should, however, be cautious with one-sided conclusions based on absolute dates. Jar burial sites like Renah Kemumu are rooted in a tradition which outside of Sumatra, i.e., Niah in Sarawak, goes back to the late Neolithic period around 1500 BCE. In Muara Payang (fig. 3-5) and Padang Sepan (at the foothills north of Bengkulu) polished stone adzes have been found in close vicinity of burial sites, therefore the Neolithic tradition seems to be almost tangible at these places. Last but not least, the already mentioned excavation of the French-Indonesian team at Benua Keling in Pasemah has shown how a Neolithic site changed its appearance to a Paleometallic site before it was finally converted into a large burial mound complex (Guillaud et al., infra).

Fig. 3-5: Two polished stone adzes from Muara Payang, Lahat (printed with courtesy of Balai Arkeologi Palembang)

What follows is that jar burial sites can be related to Neolithic traditions and that, concerning the nomenclature, it only depends on the point of view whether to give prevalence to the either Neolithic traditions or Metallic innovations at a particular site. The decision, however, is in itself problematic since it is often ideological. The terminology should not be completely based on the evaluation of material culture which in the Sumatran case seems to be a problematic instrument for the division into periods. Jar burial sites on Sumatra can not be classified as Neolithic assemblages, but they might have been still embedded in a context of Neolithisation which could have been a long lasting process.

Megalithic sites

Megaliths are by far the most well known archaeological remains of Sumatra. In the highlands they are clustered over four main regions: The Batak-region, the Minangkabau-area, the highlands of Jambi, including the Kerinci region, and Pasemah.[11] Each region has its own distinct type of megalithic setting. Thus, there is no reason to believe that one "megalithic culture" has directly influenced the other. Excavations at megalithic sites are still very rare. They prove, however, that none of the megalithic complexes dates to earlier than the first millennium CE, a fact which was formerly only indicated by the iconographic evidence of the Pasemah megaliths. Only one site in the Minangkabau region has yielded, though enigmatically, evidence for an earlier date. In Bawah Parit in the Mahat valley Indonesian archaeologist have dug up human skeletons buried under two *kris*-hilt shaped stones with carved decoration (Miksic 1986, 2004). The analysis of the bone material provided a ^{14}C date of 2,070-2,130 BP (Aziz/Siregar 1997: 20). This early date and the exceptional use of the megaliths as tombstones are difficult to explain. Also elusive is the explanation for several polished stone adzes which were found beside, not under the stones, apparently in conjunction with pottery of recent date (Miksic 2004: 196). Can the stone adzes, the burials, and the upright stones be related in a chronological and functional way, or do they appear next to each other coincidently?

The bulk of the other megaliths in the Mahat and Sinamar valley date certainly much later and overlap with the time when king Adityavarman took control over the region in the 14th century (Miksic 1987, 2004). That

[11] Concerning the megaliths in Pasemah, the highlands of Jambi and the Minangkabau area see also the contributions of Guillaud et al., infra and Tjoa-Bonatz, infra and Bakels, infra.

means that connections with the historical period of Srivijaya-Malayu in the lowlands can be claimed.[12] The same can be suggested for the megalithic sites in Pasemah and those documented in the highlands of Jambi.

Fig. 3-6: Excavated area with postholes of a house next to the megalith at Pondok, Kerinci

Two excavated sites in Pondok in the Kerinci region (fig. 3-6) and Bukit Batu Larung near Renah Kemumu in the Serampas region (fig. 3-7) show that a single megalith (*batu larung*) was erected in the centre of a settlement place and in conjunction with a stilted house which could be interpreted either as a domestic building or a meeting hall (Bonatz 2006: 317-322, fig. 29.11-29.13; Bonatz et al. 2006: 495, figs. 1-2; 497-500, fig. 6-7). Both sites have yielded a rich material culture including imports of Chinese porcelain, Indo-Pacific glass beads, and iron tools (Bonatz 2006: 318-322, fig. 29.14-29.16; Bonatz et al. 2006: 499-500). These finds prove exchange contacts with the harbour sites on the east and probably also the west coast of Sumatra. Given the date of the megalithic sites, which has been confirmed by radiocarbon dating and OSL-analysis of potsherds, trade was occurring around the 12th century CE, and thus during the period

[12] For Adityavarman's rule over the Minangkabau highland see also the contribution of Kulke, infra.

when the capital of the Srivijaya-Malayu emporium had been based at Muara Jambi. A similar connection with the lowland empires can be assumed for the Pasemah megalithic sites. Given to the Dongson-style kettledrums, metal ornaments, and weapons which are depicted on these stones, the Pasemah complex can roughly be dated between 200 BCE and 500 CE or even later.[13] That means it could at least partly coincide with the raise of Srivijaya's first city-state at Palembang in the 7th century CE (c.f. Manguin, infra).

Fig. 3-7: Megalith in the centre of the excavated area at Bukit Batu Larung, Serampas

What does this connection of megalithic sites to historical kingdoms - and thus to regions of "classical archaeology" in Indonesia - tells us about the nature of these highland sites? First, it explains why certain items such

[13] Two of these stones (Guillaud et al., infra: fig. 18-2 and 18-3) depict scenes in which bronze kettledrums, certainly one of the most prestigious goods by that time, were brought to the highlands and given to the locals. These scenes apparently mirror the ritual of exchange between the highlands and the lowlands. The phenomenon of erecting stone monuments seems to be ultimately linked to this exchange.

as metals, including prestigious bronzes like the Dongson kettledrums and the bronze flask from Kerinci,[14] Chinese porcelain, and glass beads appeared there for the first time. Second, it suggests that the erection of the megaliths could in some way have been related to the issues of trade and by extension the social-economic changes which trade caused in this region. And third, it demonstrates that at least some communities had advanced to such a prosperous level that they committed themselves to the erection of a megalith which may imply a high degree of territoriality. I would even like to go further with this statement and add that these are in fact the megaliths which give the first firm evidence for sedentism in the highlands, even if this conclusion does not necessarily imply agriculture.

It is doubtful if the subsistence of the people who erected the megaliths was already fully based on farming, or if it was still in the process of transition from foraging to farming. I raise this question because I would like to stress that material culture is only the secondary trait of the phenomenon called the Neolithic while subsistence is the primary trait of this phenomenon. In the case of highland Sumatra one should not inevitably classify a site as post-Neolithic, i.e., Metallic or early Metallic, because some metal finds have been made there. In specific historical situations one should rather expect that different material objects which are considered to be typical for certain archaeological periods such as Paleolithic, Neolithic or Metallic may have coexisted. Obsidian flakes, for example appear as early as the late Paleolithic in Sumatra but they continued to be used there without greater typological variation until modern times. The designation of highland Neolithic communities, therefore, does not exclude the possibility that megaliths and metal belonged to their specific cultural traits as did pottery and stone implements. Once again it must be said that the definition of the Neolithic does not work on the basis of material traits but only in correspondence with the reconstruction of the subsistence strategies of past communities.

Prehistoric Subsistence Strategies in the Highlands

Two palynological studies conducted in Kerinci suggest that forest clearance for swidden agriculture started around 4,000 BP (Morley 1982) or 7,000 BP (Flenley/Butler 2001), although these conclusions remain

[14] A bronze fragment, what appears to have been part of a kettledrum, and the famous bronze flask from south of lake Kerinci were already recorded in the 1930s; see van der Hoop (1940: 2001, pl. 80.1) and Wolters (1967: 60-61) on the kettledrum fragment; Glover (1990: 218-220, 224, fig. 1, pl. 5) and Glover (2004, 2008) on the bronze flask.

speculative. Palynological studies in northern Sumatra, i.e., in the Toba Lake area lead to the general conclusion that humans have been clearing forest in the upland areas for some 7,000 years or more (Flenley 1988). At first this was for shifting cultivation and at a level which allowed complete forest regeneration (Flenely 1988: 187). Only from about 2,000 BP, do permanent clearings begin to be made. A possible tree cultivation phase, attested to so far only by the pollen record of *Arenga*, the sugar palm, is in turn succeeded by today's intensive rice cultivation.

In my opinion the environmental studies, as limited as they are, prove that early forest disturbance was not necessarily caused by farming but by foraging, which included such activities as controlled burning of vegetation, gathering, and protective tending. Only the evidence for permanent forest clearance 2,000 years ago indicates the transition to a true agro-ecosystem, i.e., the process of Neolithisation.[15] We are thus envisaging a date which is not so far from that indicated by the archaeological record. It is furthermore important to note that the beginnings of farming were apparently not marked by the introduction of rice or other cereals but by vegetatively reproduced root crops, especially yams (*Discorea sp.*) and taro (*Colocasia* and *Alocasia*), which unfortunately leaves no archaeological trace. This means that the transition from foraging to food production was characterized by what David R. Harris (1973) called the "palaeotechnic tropical agriculture" in which diverse assemblages of crops and animals are raised in structural and functional interdependence, mirroring the complex structure of the natural ecosystem.[16] Shifting agriculture is the characteristic of this subsistence technique. It seems that in most parts of the Sumatran highlands this type of agriculture was practiced until very recently before fixed field agriculture, i.e., wet-rice cultivation, was introduced as an alternative source of food production.[17]

Given the possibility that sedentary communities might have practiced swidden agriculture, the status of "Neolithic" communities must be inferred from their settlement-subsistence strategies. The archaeological

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[15] For an elaborated description of the evolutionary sequence of plant cultivation see Harris (1989).

[16] See also Glover (1977: 155-158).

[17] In the Kerinci region, for example, evidence for wet-rice cultivation only goes back to Thomas Barnes' expedition in 1818 (Kathirithamby-Wells 1986). Even today in some highland regions dry rice cultivation in the upland fields appears as a modern variation of shifting agriculture since it is part of a rotational agroforestry system, i.e., rice and annuals being replaced by coffee and other crops. See Neidel (2006: 140, fn. #130 "i" on p. 408-409).

settlement sites of the highlands fit in well to the idea of communities practicing shifting or swidden agriculture because they reflect either a mobile settlement pattern, or they form a sort of communal place for people living not in villages but on their fields. Megalithic sites are the most probable candidates for such sites, but other settlements fall into this category as well.[18] A high residential mobility would then still have been an important factor for the economic orientation of the people who we designate farmers. Therefore, I propose to consider the Neolithic in the highlands of Sumatra as a period of long-term changes in economic and socio-politic complexity. Such changes include the adaptation of flexible cultivation strategies, an at least semi-sedentary lifestyle, and a prolific material culture. The circumstances under which the process of Neolithisation has been accomplished in the highlands still lay beyond our scopes. But it may be suggested that it well extends to the time when early state polities became established in the low- and midlands.

Conclusion

Considering both the archaeological and the environmental evidence discussed so far, there is no reason to believe that the beginning of the Neolithic in the Sumatran highlands was marked by the introduction of fixed field agriculture. Instead, it seems that the different regions of the highlands independently developed shifting or swidden agriculture on the basis of indigenous plants. Sedentary life-style, then, was adopted up to the point that constant movements within a broader regional ecosystem still allowed flexible subsistence strategies. These strategies probably included early farming and horticulture as well as hunting and collecting.

Based on a mobile settlement pattern, first complex settlement systems might not have started much earlier than the beginning of the first millennium BCE, and it even appears that in some regions like Serampas they only developed at the end of this millennium. Assuming that the transition to food production in combination with first steps towards sedentary life correlate with the process of Neolithisation, the main question is not why this process started so late but why it occurred at all. The answer to this question, as speculative as at it might be at the moment, lies in the archaeological evidence. All the regions in question share from a certain moment onwards, which varies within a time span of about 500 years, the same basic features of permanent settlement activities such as the marking of territories and settlement places by burial grounds and

[18] See Neidel, infra with references.

megaliths, the accumulation of local earthenwares and stone implements at certain places, and in some instances the appearance of metals, beads, and import ceramics at the same places. The latter group of finds, as it already has been said, put the highland settlements in a trade or at least economic exchange relation with the low- and midland polities of Srivijaya-Malayu.

The "pull" model, which is one of the models discussed at the beginning of this paper, seems to me plausible enough to explain the way in which the archaeological evidence relates to the definition of the Neolithic and the question of its occurrence in the highlands. In contrast, no evidence exists for stress factors caused by exceeding resources or drastic climatic changes which would have pushed the people into agriculture. And no "competitive feasting" or surplus production can be inferred from the archaeological data as to explain the investments in food production. Instead, one may reasonably assume that the protection of indigenous plants which were continuously exploited not only contributed to their genetic manipulation but also stimulated humans to cultivate them. In addition to this basic argument of a "pull" model I would like to argue in favour of a situation in which economic exchanges with the lowlands increased the reliance of the highland populations on specific resources, for example their tradable forest products, and thus pulled them into agriculture. Since trade has always been something to do with territoriality, i.e., the claim on territories from which trade goods stem or through which they are traded, it can be proposed as one of the main causes for the orientation towards sedentary life. The "pull" model may therefore not only put emphasis on economic, material, and social changes caused by external relations, it could also help to accept that the formation of early agricultural societies and early state-organized societies could have been intertwined and that therefore, in the case of Sumatra, a Neolithic Period in the highlands could very well have coexisted with a Classical Period in the coastal areas. An unlinear approach to models of cultural evolution in Sumatra is for that reasons just as hazardous as for other regions in the Southeast Asian archipelago.

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